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Economics | (12th Edition)

Step-by-step solution

Step 1 of 10

Average product – Average product is obtained by dividing the total product with the number of workers employed.

$$\text{Average Product} = \frac{\text{Total Product}}{\text{Number of Workers}}$$

[Comment](#)

Step 2 of 10

Following table shows Sue's Surfboards' Total Product schedule –

[Comment](#)

Step 3 of 10

Labor	Output
(Workers per week)	(Surfboards per week)
1	30
2	70
3	120
4	160
5	190
6	210
7	220

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Step 4 of 10

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Chapter 11, Problem 3SPA

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per week, so average product of Sue Surfboards when one worker is employed is as follows –

$$\text{Average Product} = \frac{\text{Total Product}}{\text{Number of Workers}}$$

$$\text{Average Product} = \frac{30}{1}$$

Average Product = 30 surfboards per week

When Sue Surfboard employs 2 worker per week, its total product is 70 surfboards per week, so average product of Sue Surfboards when one worker is employed is as follows –

$$\text{Average Product} = \frac{\text{Total Product}}{\text{Number of Workers}}$$

$$\text{Average Product} = \frac{70}{2}$$

Average Product = 35 surfboards per week

When Sue Surfboard employs 3 worker per week, its total product is 120 surfboards per week, so average product of Sue Surfboards when one worker is employed is as follows –

$$\text{Average Product} = \frac{\text{Total Product}}{\text{Number of Workers}}$$

$$\text{Average Product} = \frac{120}{3}$$

Average Product = 40 surfboards per week

When Sue Surfboard employs 4 worker per week, its total product is 160 surfboards per week, so average product of Sue Surfboards when one worker is employed is as follows –

$$\text{Average Product} = \frac{\text{Total Product}}{\text{Number of Workers}}$$

$$\text{Average Product} = \frac{160}{4}$$

Average Product = 40 surfboards per week

When Sue Surfboard employs 5 worker per week, its total product is 190 surfboards per week, so average product of Sue Surfboards when one worker is employed is as follows –

$$\text{Average Product} = \frac{\text{Total Product}}{\text{Number of Workers}}$$

$$\text{Average Product} = \frac{190}{5}$$

Average Product = 38 surfboards per week

When Sue Surfboard employs 6 worker per week, its total product is 210 surfboards per week, so average product of Sue Surfboards when one worker is employed is as follows –

$$\text{Average Product} = \frac{\text{Total Product}}{\text{Number of Workers}}$$

$$\text{Average Product} = \frac{210}{6}$$

Average Product = 35 surfboards per week

[Comment](#)

Step 5 of 10

When Sue Surfboard employs 7 worker per week, its total product is 220 surfboards per week, so average product of Sue Surfboards when one worker is employed is as follows –

$$\text{Average Product} = \frac{\text{Total Product}}{\text{Number of Workers}}$$



Chapter 11, Problem 3SPA

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[Comment](#)**Step 6 of 10**

Following is the average product schedule of the Sue Surfboards –

[Comment](#)**Step 7 of 10**

Labor	Total Product	Average Product
(workers per week)	(surfboards per week)	(surfboards per week)
1	30	30
2	70	35
3	120	40
4	160	40
5	190	38
6	210	35
7	220	31

[Comment](#)**Step 8 of 10**

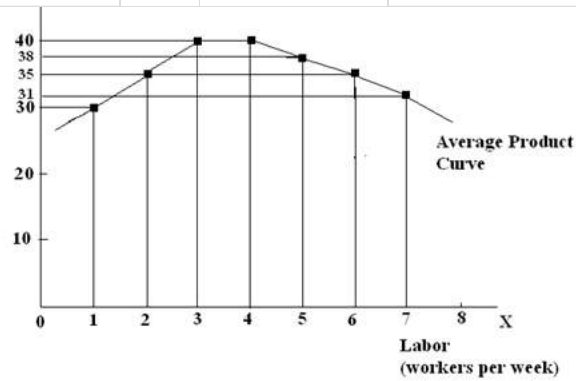
Following average product curve is based on the above mentioned average product schedule –

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Chapter 11, Problem 3SPA

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Step 10 of 10

In the above graph, labor is taken on the X – axis and output is taken on the Y – axis. The average product curve is of inverted U shape. This implies that average product curve first rises, reaches it's maximum and then falls.

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Chapter 11, Solution 8SPA

Following Table shows the Total Cost, Total Variable Cost, and Total Fixed Cost of Sue Surfboards at each level of output –

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Chapter 11, Solution 12SPA

The following table shows the production function of Jackie's Canoe Rides –

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Chapter 11, Problem 3SPA

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Economics | (12th Edition)

Step-by-step solution

Step 1 of 13

Marginal Product – Marginal product refers to the addition to the total product when one more unit of a factor of production is employed.

Alternatively,

$$\text{Marginal Product} = \frac{\text{Increase in Total Product}}{\text{One unit increase in the quantity of factor employed}}$$

[Comment](#)

Step 2 of 13

Following table shows Sue's Surfboards' total product schedule –

[Comment](#)

Step 3 of 13

Labor	Output
(Workers per week)	(Surfboards per week)
1	30
2	70
3	120
4	160
5	190
6	210
7	220

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when Sue Surfboards increase its workforce by 1 more worker, its total product increases by 40 surfboards per week. Therefore, marginal product of Sue Surfboards, when 2nd worker is employed would be,

$$\text{Marginal Product} = \frac{\text{Increase in Total Product}}{\text{One unit increase in the quantity of labor employed}}$$

$$\text{Marginal Product} = \frac{40}{1}$$

Marginal product = 40 surfboards per week

[Comment](#)

Step 5 of 13

When Sue Surfboards employs 2 workers, its total product is 70 surfboards per week. When Sue Surfboards employs 3 workers, its total product is 120 surfboards per week. So, when Sue Surfboards increase its workforce by 1 more worker, its total product increases by 50 surfboards per week. Therefore, marginal product of Sue Surfboards, when 3rd worker is employed would be,

$$\text{Marginal Product} = \frac{\text{Increase in Total Product}}{\text{One unit increase in the quantity of labor employed}}$$

$$\text{Marginal Product} = \frac{50}{1}$$

Marginal product = 50 surfboards per week

[Comment](#)

Step 6 of 13

When Sue Surfboards employs 3 workers, its total product is 120 surfboards per week. When Sue Surfboards employs 4 workers, its total product is 160 surfboards per week. So, when Sue Surfboards increase its workforce by 1 more worker, its total product increases by 40 surfboards per week. Therefore, marginal product of Sue Surfboards, when 4th worker is employed would be,

$$\text{Marginal Product} = \frac{\text{Increase in Total Product}}{\text{One unit increase in the quantity of labor employed}}$$

$$\text{Marginal Product} = \frac{40}{1}$$

Marginal product = 40 surfboards per week

[Comment](#)

Step 7 of 13

When Sue Surfboards employs 4 workers, its total product is 160 surfboards per week. When Sue Surfboards employs 5 workers, its total product is 190 surfboards per week. So, when Sue Surfboards increase its workforce by 1 more worker, its total product increases by 30 surfboards per week. Therefore, marginal product of Sue Surfboards, when 5th worker is employed would be,

$$\text{Marginal Product} = \frac{\text{Increase in Total Product}}{\text{One unit increase in the quantity of labor employed}}$$

$$\text{Marginal Product} = \frac{30}{1}$$

Step 8 of 13

When Sue Surfboards employs 5 workers, its total product is 190 surfboards per week. When Sue Surfboards employs 6 workers, its total product is 210 surfboards per week. So, when Sue Surfboards increase its workforce by 1 more worker, its total product increases by 20 surfboards per week. Therefore, marginal product of Sue Surfboards, when 6th worker is employed would be,

MarginalProduct = $\frac{\text{Increase in Total Product}}{\text{One unit increase in the quantity of labor employed}}$

MarginalProduct = $\frac{20}{1}$

Marginal product = 20 surfboards per week

[Comment](#)

Step 9 of 13

When Sue Surfboards employs 6 workers, its total product is 210 surfboards per week. When Sue Surfboards employs 7 workers, its total product is 220 surfboards per week. So, when Sue Surfboards increase its workforce by 1 more worker, its total product increases by 10 surfboards per week. Therefore, marginal product of Sue Surfboards, when 7th worker is employed would be,

MarginalProduct = $\frac{\text{Increase in Total Product}}{\text{One unit increase in the quantity of labor employed}}$

MarginalProduct = $\frac{10}{1}$

Marginal product = 10 surfboards per week

Following is the Marginal Product schedule of the Sue Surfboards –

[Comment](#)

Step 10 of 13

Labor	Total product	Marginal Product
(workers per week)	(surfboards per week)	(surfboards per week)
1	30	
		40
2	70	
		50



Chapter 11, Problem 4SPA

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4	160	
		30
5	190	
		20
6	210	
		10
7	220	

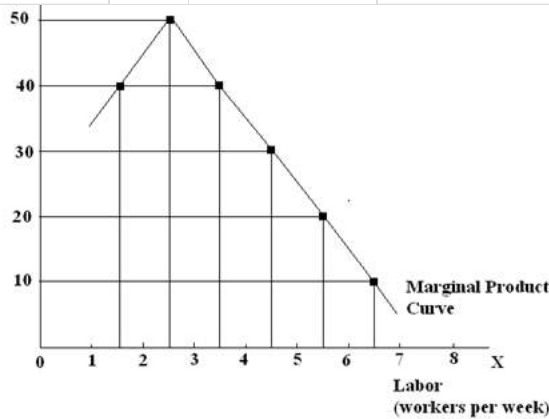
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Step 11 of 13

Following marginal product curve is based on the above mentioned marginal product schedule –

[Comment](#)

Step 12 of 13



[Comment](#)

Step 13 of 13

In the above graph, labor is taken on the X – axis and output is taken on the Y – axis. The above drawn marginal product curve is plotted at the mid – points. This implies that at the quantity of 2.5 workers, the marginal product is 50 surfboards per week.

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Chapter 11, Solution 8SPA

Following Table shows the Total Cost, Total Variable Cost, and Total Fixed Cost of Sue Surfboards at each level of output –

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Chapter 11, Solution 12SPA

The following table shows the production function of Jackie's Canoe Rides –

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Step-by-step solution

Step 1 of 5

Following table shows Sue's Surfboards' Total product, Average product, and Marginal product schedule –

[Comment](#)

Step 2 of 5

Labor	Total Product	Average Product	Marginal Product
(workers per week)	(surfboards per week)	(surfboards per week)	(surfboards per week)
1	30	30	
			40
2	70	35	
			50
3	120	40	
			40
4	160	40	
			30
5	190	38	
			20
6	210	35	
			10
7	220	31	

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Step 3 of 5

(a) A firm experience increasing marginal returns due to increased specialization and division of labor. When increasing marginal returns occur, the marginal product of each successive worker employed exceeds the marginal product of the previous worker employed. So, a firm would enjoy the benefit of increased specialization and division of labor over the range of output in which the marginal product of labor is increasing.

As above table indicates that up to the employment of 3 workers marginal product of labor is increasing. So, Sue's surfboards will be enjoying the benefits of increased specialization and division of labor until it employs 3 workers per week and produce 120 surfboards per week.

[Comment](#)

Step 4 of 5

(b) A firm experiences the diminishing marginal returns when marginal product of each successive worker employed is less than the marginal product of the previous worker employed. Alternatively, a firm will experience diminishing marginal returns over that range of output during which marginal product of labor is decreasing.

As above table indicates that after the employment of third worker, marginal product has started decreasing. So, Sue's surfboard will experience diminishing marginal product of labor when it employs more than 3 workers and produce more than 120 surfboards per week.

[Comment](#)

Step 5 of 5

(c) The output range over which the Sue Surfboards experience an increasing average product of labor but a diminishing marginal product of labor is when it produces between 70 to 160 surfboards per week and employs between 2 to 4 workers per week.

As above table indicates that when Sue's surfboard increases the number of workers from 2 to 3, and eventually to 4, its average product of labor still increases but its marginal product of labor starts declining.

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Chapter 11, Solution 8SPA

Following Table shows the Total Cost, Total Variable Cost, and Total Fixed Cost of Sue Surfboards at each level of output –

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Chapter 11, Solution 12SPA

The following table shows the production function of Jackie's Canoe Rides –

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Step-by-step solution

Step 1 of 1

Average product increases as long as marginal product is greater than the average product. In other words, when $MP > AP$, AP rises. So, even though marginal product is decreasing but if it still remains greater than the average product then due to the above stated relationship between the two average product will increase.

This implies that, when firm hires an additional worker, he may be producing less than previously hired worker but still his contribution is more than the average contribution of each worker.

Thus, it is possible for a firm to experience simultaneously an increasing average product but a diminishing marginal product.

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Chapter 11, Solution 8SPA

Following Table shows the Total Cost, Total Variable Cost, and Total Fixed Cost of Sue Surfboards at each level of output –

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The following table shows the production function of Jackie's Canoe Rides –

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Chapter 11, Problem 7SPA

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Step-by-step solution

Step 1 of 5

Total fixed cost – It is the cost of the firm's fixed factors. The quantities of fixed factors do not change as output changes, so total fixed cost is the same at all outputs.

It is given that Sue Surfboards total fixed cost is \$1,000 a week.

Total variable cost – It is the cost of firm's variable factors. For Sue Surfboards, labor is the variable factor, so this component of cost is its wage bill. Total variable cost changes as output changes.

It is given that Sue Surfboards hires workers at \$500 per week. So, if it hires 1 worker its total variable cost will be \$500 ($\500×1). If it increase its labor force and employs one more worker, now it has to workers its total variable cost will increase to

\$1,000 ($\500×2).

Total cost – It is the cost of all the factors of production it uses. Total cost is the sum of total fixed cost and total variable cost. That is,

$$TC = TFC + TVC$$

Following table shows the Total Cost, Total Variable Cost, and Total Fixed Cost of each level of output –

[Comment](#)

Step 2 of 5

Labor	Output	Total Fixed Cost	Total Variable Cost	Total Cost (TFC + TVC)
(workers per week)	(surfboards per week)	(dollars per week)	(dollars per week)	(dollars per week)
1	30	\$1,000	\$500	\$1,500
2	70	1,000	1,000	2,000
3	120	1,000	1,500	2,500
4	160	1,000	2,000	3,000
5	190	1,000	2,500	3,500
6	210	1,000	3,000	4,000
7	220	1,000	3,500	4,500

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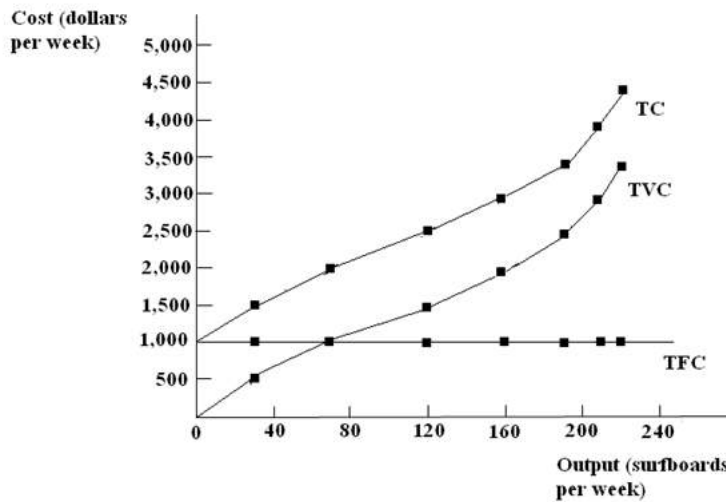
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Step 3 of 5

Following Total Cost, Total Variable cost, and Total Fixed cost curves are based on above mentioned table –

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Step 4 of 5



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Step 5 of 5

In the above figure, on X – axis Output (surfboards per week) has been taken while on Y – axis Cost (dollars per week) has been taken.

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Chapter 11, Solution 8SPA

Following Table shows the Total Cost, Total Variable Cost, and Total Fixed Cost of Sue Surfboards at each level of output –

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Chapter 11, Solution 12SPA

The following table shows the production function of Jackie's Canoe Rides –

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Step-by-step solution

Step 1 of 39

Following Table shows the Total Cost, Total Variable Cost, and Total Fixed Cost of Sue Surfboards at each level of output –

[Comment](#)

Step 2 of 39

Output	Total Fixed Cost	Total Variable Cost	Total Cost (TFC + TVC)
(surfboards per week)	(dollars per week)	(dollars per week)	(dollars per week)
30	\$1,000	\$500	\$1,500
70	1,000	1,000	2,000
120	1,000	1,500	2,500
160	1,000	2,000	3,000
190	1,000	2,500	3,500
210	1,000	3,000	4,000
220	1,000	3,500	4,500

[Comment](#)

Step 3 of 39

Average fixed cost – Average fixed cost is the Total Fixed Cost per unit of output.

Alternatively,

$$AFC = \frac{TFC}{Q}$$

Where,

AFC = Average fixed cost

TFC = Total fixed cost

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Step 4 of 39

For example, When Sue Surfboards produce 30 surfboards; its total fixed cost is \$1,000, so average fixed cost would be,

$$AFC = \frac{TFC}{Q}$$

$$AFC = \frac{1,000}{30}$$

AFC = \$33.33 per surfboard

[Comment](#)

Step 5 of 39

When Sue Surfboards produce 70 surfboards; its total fixed cost is \$1,000, so average fixed cost would be,

$$AFC = \frac{TFC}{Q}$$

$$AFC = \frac{1,000}{70}$$

AFC = \$14.28 per surfboard

[Comment](#)

Step 6 of 39

When Sue Surfboards produce 120 surfboards; its total fixed cost is \$1,000, so average fixed cost would be,

$$AFC = \frac{TFC}{Q}$$

$$AFC = \frac{1,000}{120}$$

AFC = \$8.33 per surfboard

[Comment](#)

Step 7 of 39

When Sue Surfboards produce 160 surfboards; its total fixed cost is \$1,000, so average fixed cost would be,

$$AFC = \frac{TFC}{Q}$$

$$AFC = \frac{1,000}{160}$$

AFC = \$6.25 per surfboard

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$$AFC = \frac{TFC}{Q}$$

$$AFC = \frac{1,000}{190}$$

AFC = \$5.26 per surfboard

[Comment](#)

Step 9 of 39

When Sue Surfboards produce 210 surfboards; its total fixed cost is \$1,000, so average fixed cost would be,

$$AFC = \frac{TFC}{Q}$$

$$AFC = \frac{1,000}{210}$$

AFC = \$4.76 per surfboard

[Comment](#)

Step 10 of 39

When Sue Surfboards produce 220 surfboards; its total fixed cost is \$1,000, so average fixed cost would be,

$$AFC = \frac{TFC}{Q}$$

$$AFC = \frac{1,000}{220}$$

[Comment](#)

Step 11 of 39

AFC = \$4.54 per surfboard

Average variable cost – Average variable cost is the total variable cost per unit of output.

Alternatively,

$$AVC = \frac{TVC}{Q}$$

Where,

AVC = Average variable cost

TVC = Total variable cost

Q = Quantity produced

[Comment](#)

Step 12 of 39

For example, When Sue Surfboards produce 30 surfboards; its total variable cost is \$500, so average variable cost would be,

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Step 13 of 39

When Sue Surfboards produce 70 surfboards; its total variable cost is \$1,000, so average variable cost would be,

$$AVC = \frac{TVC}{Q}$$

$$AVC = \frac{\$1,000}{70}$$

AVC = \$14.28 per surfboard

[Comment](#)

Step 14 of 39

When Sue Surfboards produce 70 surfboards; its total variable cost is \$1,000, so average variable cost would be,

$$AVC = \frac{TVC}{Q}$$

$$AVC = \frac{\$1,000}{70}$$

AVC = \$14.28 per surfboard

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Step 15 of 39

When Sue Surfboards produce 120 surfboards; its total variable cost is \$1,500, so average variable cost would be,

$$AVC = \frac{TVC}{Q}$$

$$AVC = \frac{\$1,500}{120}$$

AVC = \$12.5 per surfboard

[Comment](#)

Step 16 of 39

When Sue Surfboards produce 160 surfboards; its total variable cost is \$2,000, so average variable cost would be,

$$AVC = \frac{TVC}{Q}$$

$$AVC = \frac{\$2,000}{160}$$

AVC = \$12.5 per surfboard

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When Sue Surfboards produce 190 surfboards; its total variable cost is \$2,500, so average variable cost would be,

$$AVC = \frac{TVC}{Q}$$

$$AVC = \frac{\$2,500}{190}$$

AVC = \$13.16 per surfboard

[Comment](#)

Step 18 of 39

When Sue Surfboards produce 210 surfboards; its total variable cost is \$3,000, so average variable cost would be,

$$AVC = \frac{TVC}{Q}$$

$$AVC = \frac{\$3,000}{210}$$

AVC = \$14.28 per surfboard

[Comment](#)

Step 19 of 39

When Sue Surfboards produce 220 surfboards; its total variable cost is \$3,500, so average variable cost would be,

$$AVC = \frac{TVC}{Q}$$

$$AVC = \frac{\$3,500}{220}$$

AVC = \$15.91 per surfboard

[Comment](#)

Step 20 of 39

Average total cost – Average total cost is the total cost per unit of output.

Average total cost is also the sum of Average variable cost and Average fixed cost. That is,

$$ATC = AFC + AVC$$

For example, When Sue Surfboards produce 30 surfboards; its average variable cost (AVC) is \$33.33 per surfboard and average fixed cost (AFC) is \$16.67 per surfboard.

So, average total cost (ATC) would be,

$$ATC = AVC + AFC$$

$$ATC = \$33.33 + \$16.67$$

$$= \$50.00 \text{ per surfboard}$$

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So, average total cost (ATC) would be,

$$ATC = AVC + AFC$$

$$ATC = \$14.28 + \$14.28$$

$$= \$28.56 \text{ per surfboard}$$

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Step 22 of 39

When Sue Surfboards produce 120 surfboards; its average variable cost (AVC) is \$8.33 per surfboard and average fixed cost (AFC) is \$12.50 per surfboard.

So, average total cost (ATC) would be,

$$ATC = AVC + AFC$$

$$ATC = \$8.33 + \$12.50$$

$$= \$20.83 \text{ per surfboard}$$

[Comment](#)

Step 23 of 39

When Sue Surfboards produce 160 surfboards; its average variable cost (AVC) is \$6.25 per surfboard and average fixed cost (AFC) is \$12.50 per surfboard.

So, average total cost (ATC) would be,

$$ATC = AVC + AFC$$

$$ATC = \$6.25 + \$12.50$$

$$= \$18.75 \text{ per surfboard}$$

[Comment](#)

Step 24 of 39

When Sue Surfboards produce 190 surfboards; its average variable cost (AVC) is \$5.26 per surfboard and average fixed cost (AFC) is \$13.16 per surfboard.

So, average total cost (ATC) would be,

$$ATC = AVC + AFC$$

$$ATC = \$5.26 + \$13.16$$

$$= \$18.42 \text{ per surfboard}$$

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Step 25 of 39

When Sue Surfboards produce 210 surfboards; its average variable cost (AVC) is \$4.76 per surfboard and average fixed cost (AFC) is \$14.28 per surfboard.

So, average total cost (ATC) would be,

$$ATC = AVC + AFC$$

$$ATC = \$4.76 + \$14.28$$

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Step 26 of 39

When Sue Surfboards produce 220 surfboards; its average variable cost (AVC) is \$4.54 per surfboard and average fixed cost (AFC) is \$15.91 per surfboard.

So, average total cost (ATC) would be,

$$ATC = AVC + AFC$$

$$ATC = \$4.54 + \$15.91$$

$$= \$20.46 \text{ per surfboard}$$

[Comment](#)

Step 27 of 39

Marginal cost – Marginal cost is the increase in total cost when one more unit of output is produced.

Alternatively,

$$\text{Marginal Cost} = \frac{\text{Increase in Total Cost}}{\text{Increase in Total output}}$$

[Comment](#)

Step 28 of 39

For example, When Sue Surfboards produce 30 surfboards; its total cost is \$1,500 per week. When it Sue Surfboards produce 70 surfboards: its total cost is \$2,000 per week. So, when Sue Surfboards increase its output by 40 surfboards, its total cost increases by \$500 per week. So, its marginal cost would be,

$$\text{Marginal Cost} = \frac{\text{Increase in Total Cost}}{\text{Increase in Total output}}$$

$$\text{Marginal Cost} = \frac{\$500}{40}$$

$$\text{Marginal cost} = 12.50 \text{ per surfboard}$$

[Comment](#)

Step 29 of 39

When Sue Surfboards produce 70 surfboards; its total cost is \$2,000 per week. When it Sue Surfboards produce 120 surfboards: its total cost is \$2,500 per week. So, when Sue Surfboards increase its output by 50 surfboards, its total cost increases by \$500 per week. So, its marginal cost would be,

$$\text{Marginal Cost} = \frac{\text{Increase in Total Cost}}{\text{Increase in Total output}}$$

$$\text{Marginal Cost} = \frac{\$500}{50}$$

$$\text{Marginal cost} = 10.00 \text{ per surfboard}$$

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Chapter 11, Problem 8SPA	Bookmark	Show all steps: ON
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increase its output by 40 surfboards, its total cost increases by \$500 per week. So, its marginal cost would be,

$$\text{Marginal Cost} = \frac{\text{Increase in Total Cost}}{\text{Increase in Total output}}$$

$$\text{Marginal Cost} = \frac{\$500}{40}$$

Marginal cost = 12.50 per surfboard

[Comment](#)

Step 31 of 39

When Sue Surfboards produce 160 surfboards; its total cost is \$3,000 per week. When it Sue Surfboards produce 190 surfboards: its total cost is \$3,500 per week. So, when Sue Surfboards increase its output by 30 surfboards, its total cost increases by \$500 per week. So, its marginal cost would be,

$$\text{Marginal Cost} = \frac{\text{Increase in Total Cost}}{\text{Increase in Total output}}$$

$$\text{Marginal Cost} = \frac{\$500}{30}$$

Marginal cost = 16.67 per surfboard

[Comment](#)

Step 32 of 39

When Sue Surfboards produce 190 surfboards; its total cost is \$3,500 per week. When it Sue Surfboards produce 210 surfboards: its total cost is \$4,000 per week. So, when Sue Surfboards increase its output by 20 surfboards, its total cost increases by \$500 per week. So, its marginal cost would be,

$$\text{Marginal Cost} = \frac{\text{Increase in Total Cost}}{\text{Increase in Total output}}$$

$$\text{Marginal Cost} = \frac{\$500}{20}$$

Marginal cost = 25 per surfboard

[Comment](#)

Step 33 of 39

When Sue Surfboards produce 210 surfboards; its total cost is \$4,000 per week. When it Sue Surfboards produce 220 surfboards: its total cost is \$4,500 per week. So, when Sue Surfboards increase its output by 10 surfboards, its total cost increases by \$500 per week. So, its marginal cost would be,

$$\text{Marginal Cost} = \frac{\text{Increase in Total Cost}}{\text{Increase in Total output}}$$

$$\text{Marginal Cost} = \frac{\$500}{10}$$

Marginal cost = 50 per surfboard

[Comment](#)

Chapter 11, Problem 8SPA	Bookmark	Show all steps: ☒ ON
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of Sue Surfboards at each level of output –

[Comment](#)

Step 35 of 39

Output	AFC	AVC	ATC	MC
(surfboards per week)	(dollars per surfboard)	(dollars per surfboard)	(dollars per surfboard)	(dollars per surfboard)
30	33.33	16.67	50	
				12.5
70	14.28	14.28	28.58	
				10
120	8.33	12.5	20.83	
				12.5
160	6.25	12.5	18.75	
				16.67
190	5.26	13.16	18.42	
				25
210	4.76	14.28	19.04	
				50
220	4.54	15.91	20.46	

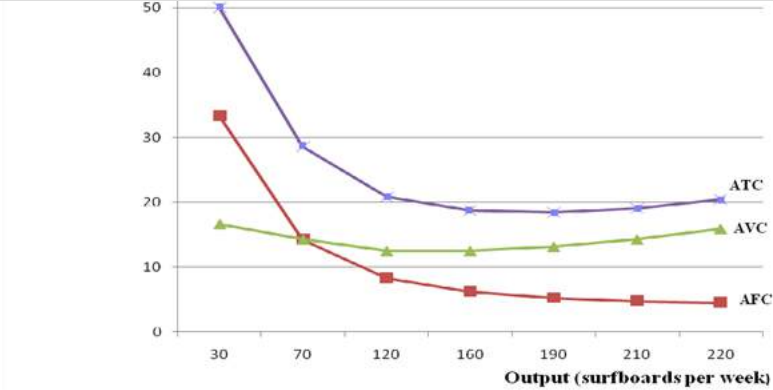
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Step 36 of 39

Following are the Short – run average cost curves of the Sue Surfboards based on above table –

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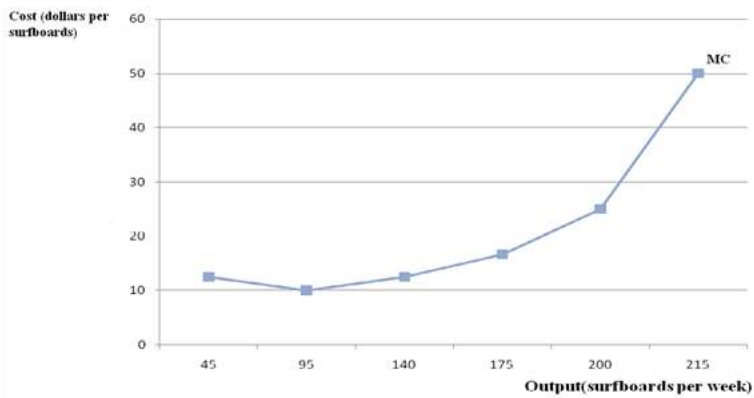
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Step 38 of 39

Following is the marginal cost curve of the Sue Surfboards based on above table –

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Step 39 of 39



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Chapter 11, Solution 8SPA

Following Table shows the Total Cost, Total Variable Cost, and Total Fixed Cost of Sue Surfboards at each level of output –

Chapter 11, Solution 12SPA

The following table shows the production function of Jackie's Canoe Rides –

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Step-by-step solution

Step 1 of 4

Number of labors and the output is known. Now in order to draw the curves some additional data is required and it can be found as shown below:

Calculation for Marginal Product and Average Product:

Labor (Workers Per Week)	Output (Surfboard Per Week)	Marginal Product (MP)	Average Product (AP)
0	0	-	-
1	30	30	30
2	70	40	35
3	120	50	40
4	160	40	40
5	190	30	38
6	210	20	35
7	220	10	31.43

[Comment](#)

Step 2 of 4

Marginal Product is calculated by subtracting the output generated by the 1st labor from the output generated by 2nd labor.

Average Product is calculated by dividing the number of labors to the output generated by the same labor.

Figure representing the 'AP' curve and 'MP' curve is given below.

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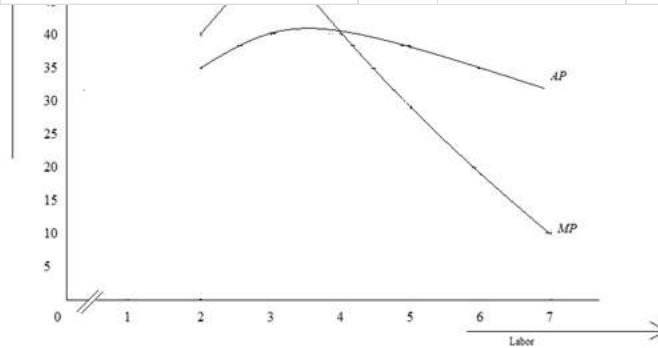
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Step 3 of 4

Table containing the values for marginal cost and average variable cost is given as follows:

Labor (Workers Per Week)	Output (Surfboard Per Week)	Fixed Cost (FC)	Variable Cost (VC)	Total Cost (TC)	Marginal Cost (MC)	Average Variable Cost (AVC)
0	0	1000	0	1000		
1	30	1000	500	1500	16.67	16.67
2	70	1000	1000	2000	12.5	14.29
3	120	1000	1500	2500	10	12.50
4	160	1000	2000	3000	12.5	12.50
5	190	1000	2500	3500	16.67	13.16
6	210	1000	3000	4000	25	14.29
7	220	1000	3500	4500	50	15.91

The formula for finding the marginal cost is as follows:

$$\text{Marginal Cost} = \frac{\Delta TC}{\Delta \text{Output}}$$

$$\text{Marginal Cost} = \frac{1500 - 1000}{30 - 0}$$

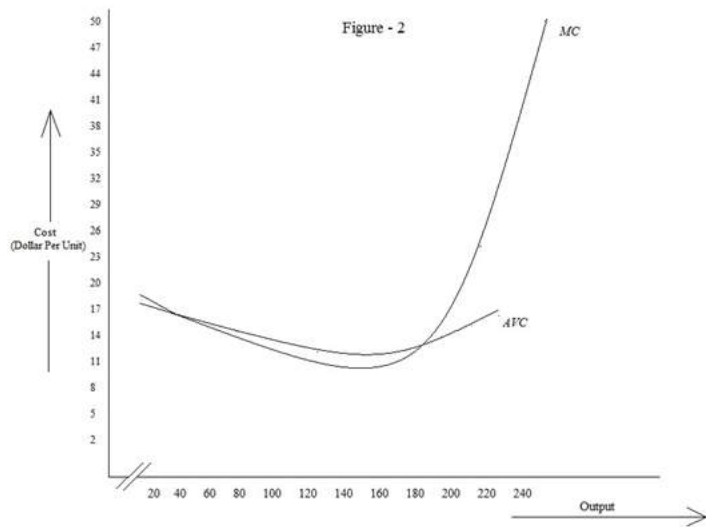
$$\text{Marginal Cost} = \frac{500}{30} = 16.67$$

Above marginal cost is found at the output level of 30 units. Similarly for the consecutive output levels, marginal cost is found.

Average Variable cost is found by division of variable cost by output.

Figure representing the 'MC' curve and 'AVC' curve is given below.

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Finding Connection in above graphs:

In figure-1 increase in worker from 2 workers to 3 workers per week, output increases to 120 surfboards per week. Marginal product and average product rises in figure -1 and marginal cost and average variable cost declines. At the point of maximum marginal product (50 units), marginal cost (\$10.00) is minimum.

As number of labor increases from 3 to 4 workers per week in figure-1, the output increases from 120 to 160 surfboards per week in figure-2. Marginal product is equal to 40 units (Figure-1) and the marginal cost start to increase at 160 units of output (Figure-2).

As number of labor increases from 4 to 5 workers per week (Figure-1), the level of output increases to 190 surfboards per week (Figure-2). Marginal product decreases and Marginal Cost increases.

At the point of maximum average product is (40 units) in figure-1, average variable cost is minimum (\$12.50) in figure-2. As the number of labor increases, output increases and average product falls while average variable cost increases.

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Following Table shows the Total Cost, Total Variable Cost, and Total Fixed Cost of Sue Surfboards at each level of output –

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- Chapter 11, Solution 12SPA

The following table shows the production function of Jackie's Canoe Rides –

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Step-by-step solution

Step 1 of 11

The following table shows the production function of Jackie's Canoe Rides –

[Comment](#)

Step 2 of 11

Labor	Output
(workers per day)	(rides per day)
	Plant 1 Plant 2
10	20 40
20	40 60
30	65 75
40	75 85
Canoes	10 20

[Comment](#)

Step 3 of 11

Rent of canoe is \$100 a day. This is the fixed cost for Jackie's Canoe Rides.

Each canoe operator is paid \$50 a day. This is the variable cost for Jackie's canoe Rides.

For example, if Jackie's operates plant 1 with 10 canoes and hires 10 workers, it undertakes 20 rides. Its total cost is as follows –

Total fixed cost = $10 \times \$100$

= \$1,000

Total variable cost = $\$50 \times 10$

= \$500

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Total cost = Total variable cost + Total fixed cost
 = \$500 + \$1,000
 = \$1,500

Average total cost is the total cost divided by quantity produced. As above calculation shows that for undertaking 20 rides, Jackie's is incurring total cost of \$1,500. So,

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Step 5 of 11

$$ATC = \frac{TC}{\text{Quantity produced}}$$

$$ATC = \frac{\$1,500}{20}$$

$$= \$75 \text{ per ride}$$

Similarly, ATC for other level of output (number of rides undertaken per day) could be calculated.

[Comment](#)

Step 6 of 11

Now, if Jackie's operates plant 2 with 20 canoes and hires 10 workers, it undertakes 40 rides. Its total cost is as follows –

Total fixed cost = 20 × \$100
 = \$2,000
 Total variable cost = \$50 × 10
 = \$500

Total cost = Total variable cost + Total fixed cost
 = \$500 + \$2,000
 = \$2,500

[Comment](#)

Step 7 of 11

Average total cost is the total cost divided by quantity produced. As above calculation shows that for undertaking 40 rides, Jackie's is incurring total cost of \$2,500. So,

$$ATC = \frac{TC}{\text{Quantity produced}}$$

$$ATC = \frac{\$2,500}{40}$$

$$= \$62.5 \text{ per ride}$$

Similarly, ATC for other level of output (number of rides undertaken per day) could be calculated.

[Comment](#)

Step 8 of 11

Following table shows the ATC when Jackie's operates Plant 1 –

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(rides per day)	(dollars per ride)
20	75
40	50
65	38.46
75	40

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Step 9 of 11

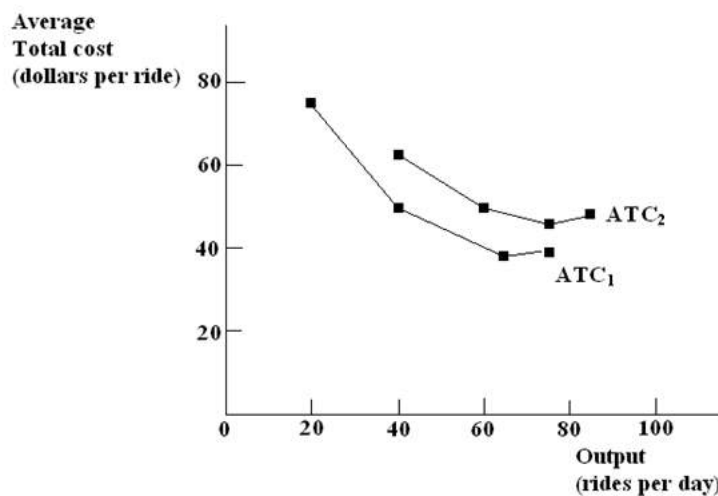
Following table shows the ATC when Jackie's operates Plant 2 –

Output	Average Total Cost (ATC)
(rides per day)	(dollars per ride)
40	62.50
60	50
75	46.67
85	47.06

[Comment](#)

Step 10 of 11

Following figure shows the average total cost curve of Jackie's canoe rides when it operate Plant 1 and Plant 2 –



Step 11 of 11

In the above figure output is taken on the X-axis and average total cost (ATC) is taken on Y – axis. ATC_1 represents average total cost when Jackie's operate Plant 1 and ATC_2 represents average total cost when Jackie's operate Plant 2.

These ATC curves differ because they both pertain to different plants. ATC_1 represents Plant 1 with 10 canoes while ATC_2 represents Plant 2 with 20 canoes.

Secondly, for each short – run ATC curve, the larger the plant, the greater is the output at which average total cost is at minimum. As above figure shows that for smaller Plant 1, ATC is minimum when 65 rides per day is conducted while for larger plant 2, ATC is minimum when 75 rides per day is conducted.

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Following Table shows the Total Cost, Total Variable Cost, and Total Fixed Cost of Sue Surfboards at each level of output –

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Chapter 11, Solution 12SPA

The following table shows the production function of Jackie's Canoe Rides –

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Step-by-step solution

Step 1 of 10

The following table shows the production function of Jackie's Canoe Rides –

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Step 2 of 10

Labor	Output
(workers per day)	(rides per day)
	Plant 3 Plant 4
10	55 65
20	75 85
30	90 100
40	100 110
Canoes	30 40

[Comment](#)

Step 3 of 10

Rent of canoe is \$100 a day. This is the fixed cost for Jackie's Canoe Rides.

Each canoe operator is paid \$50 a day. This is the variable cost for Jackie's canoe Rides.

For example, if Jackie's operates plant 3 with 30 canoes and hires 10 workers, it undertakes 55 rides. Its total cost is as follows –

Total fixed cost = $30 \times \$100$

= \$3,000

Total variable cost = $\$50 \times 10$

= \$500

Total cost = Total variable cost + Total fixed cost

= \$500 + \$3,000

= \$3,500

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Step 4 of 10

Average total cost is the total cost divided by quantity produced. As above calculation shows that for undertaking 55 rides, Jackie's is incurring total cost of \$3,500. So,

$$ATC = \frac{TC}{\text{Quantity produced}}$$

$$ATC = \frac{\$3,500}{55}$$

$$= \$63.64 \text{ per ride}$$

Similarly, ATC for other level of output (number of rides undertaken per day) could be calculated.

[Comment](#)

Step 5 of 10

Now, if Jackie's operates plant 4 with 40 canoes and hires 10 workers, it undertakes 65 rides. Its total cost is as follows –

$$\text{Total fixed cost} = 40 \times \$100$$

$$= \$4,000$$

$$\text{Total variable cost} = \$50 \times 10$$

$$= \$500$$

$$\text{Total cost} = \text{Total variable cost} + \text{Total fixed cost}$$

$$= \$500 + \$4,000$$

$$= \$4,500$$

[Comment](#)

Step 6 of 10

Average total cost is the total cost divided by quantity produced. As above calculation shows that for undertaking 65 rides, Jackie's is incurring total cost of \$4,500. So,

$$ATC = \frac{TC}{\text{Quantity produced}}$$

$$ATC = \frac{\$4,500}{65}$$

$$= \$69.23 \text{ per ride}$$

Similarly, ATC for other level of output (number of rides undertaken per day) could be calculated.

[Comment](#)

Step 7 of 10

Following table shows the ATC when Jackie's operates Plant 3 –

Output	Average Total Cost (ATC)
(rides per day)	(dollars per ride)
55	63.64

90	50
100	50

[Comment](#)

Step 8 of 10

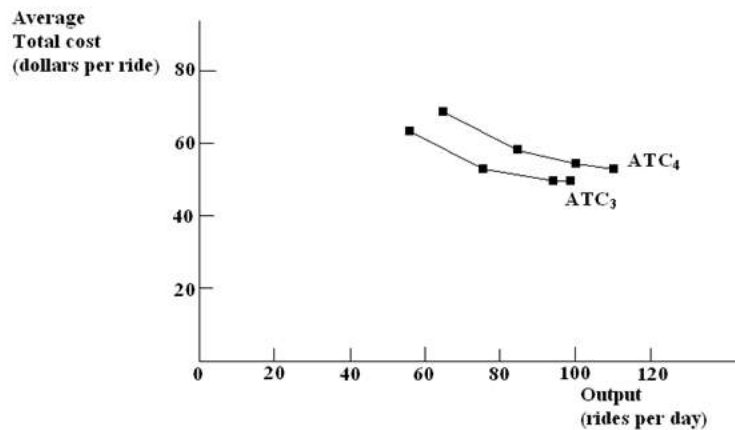
Following table shows the ATC when Jackie's operates Plant 4 –

Output	Average Total Cost (ATC)
(rides per day)	(dollars per ride)
65	69.23
85	58.82
100	55.00
110	54.55

[Comment](#)

Step 9 of 10

Following figure shows the average total cost curve of Jackie's canoe rides when it operate Plant 3 and Plant 4 –


[Comment](#)

Step 10 of 10

In the above figure output is taken on the X-axis and average total cost (ATC) is taken on Y – axis. ATC₃ represents average total cost when Jackie's operate Plant 3 and ATC₄ represents average total cost when Jackie's operate Plant 4.

Secondly, for each short – run ATC curve, the larger the plant, the greater is the output at which average total cost is at minimum. As above figure shows that for smaller Plant 3, ATC is minimum when 90 or 100 rides per day is conducted while for larger plant 2, ATC is minimum when 110 rides per day is conducted.

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Chapter 11, Solution 8SPA

Following Table shows the Total Cost, Total Variable Cost, and Total Fixed Cost of Sue Surfboards at each level of output –

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Chapter 11, Solution 12SPA

The following table shows the production function of Jackie's Canoe Rides –

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Step-by-step solution

Step 1 of 6

"LRAC curve tells the minimum efficient scale of point on which firm should operate."

In order to draw the LRAC curve, ATC curves for different output has to be drawn first. Therefore calculation for ATC curve is as follows:

For Plant 1:

Labor (workers per day)	Output (Rides Per Day For Plant 1) Q	Total Fixed Cost (TFC) In \$	AFC (=TFC/Q) In \$	Total Variable Cost (TVC) In \$	AVC (=TVC/Q) In \$	ATC ₁ (ATC= AFC+AVC) In \$
10	20	1000	50	500	25	75
20	40	1000	25	1000	25	50
30	65	1000	15	1500	23.08	38.46
40	75	1000	13.33	2000	26.67	40

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Step 2 of 6

Cost of Canoe Rides is \$100 per day per machine and the cost of labor is \$50 per day per labor. Since for first plant number of Canoes is 10 and the cost of each of the Canoe is \$100 so the fixed cost will be as follows:

Total Fixed Cost = Cost Per Canoe × Total Number of Canoes

Total Fixed Cost = 10 × (100)

Total Fixed Cost = \$1000

Similarly the Total Variable cost is as follows:

Total Variable Cost = Cost Per Canoe × Total number of Labors

Total Variable Cost = 50 × 10

Total Variable Cost = \$500

When there are 10 labors deployed in the operation, the total variable cost is coming out to be \$500 the same calculation is done for different number of labors deployed in the operation.

The calculation for AVC, AFC and ATC₁ is itself explained in the table. As a result of all the calculation data of ATC₁ at different level of output is found.

Similarly calculation for **Plant 2**, **Plant 3** and **Plant 4** is done and the data so obtained is found in the following three tables as follows:

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For Plant 2:

Labor (workers per day)	Output (Rides Per Day For Plant 2) Q	Total Fixed Cost (TFC) In \$	AFC $= \left(\frac{TFC}{Q} \right)$ In \$	Total Variable Cost (TVC) In \$	AVC $= \left(\frac{TVC}{Q} \right)$ In \$	ATC ₂ (ATC = AFC + AVC) In \$
10	40	2000	50	500	12.5	62.50
20	60	2000	33.33	1000	16.67	50
30	75	2000	26.67	1500	20	46.67
40	85	2000	23.53	2000	23.53	47.06

For Plant 3:

Labor (workers per day)	Output (Rides Per Day For Plant 3) Q	Total Fixed Cost (TFC) In \$	AFC $= \left(\frac{TFC}{Q} \right)$ In \$	Total Variable Cost (TVC) In \$	AVC $= \left(\frac{TVC}{Q} \right)$ In \$	ATC ₃ (ATC = AFC + AVC) In \$
10	55	3000	54.55	500	9.09	63.64
20	75	3000	40	1000	13.33	53.33
30	90	3000	33.33	1500	16.67	50
40	100	3000	30	2000	20	50

For Plant 4:

Labor (workers per day)	Output (Rides Per Day For Plant 4) Q	Total Fixed Cost (TFC) In \$	AFC (=TFC/Q) In \$	Total Variable Cost (TVC) In \$	AVC (=TVC/Q) In \$	ATC ₄ (ATC = AFC + AVC) In \$
10	65	4000	61.54	500	7.69	69.23
20	85	4000	47.06	1000	11.76	58.82
30	100	4000	40	1500	15	55
40	110	4000	36.36	2000	18.18	54.55

a.

C the basis of above data, ATC curves for all the plants are drawn and LRAC curve is going through the minimum point of all the ATC curves. Figure 1 is representing the graphical form of data and is shown as follows:

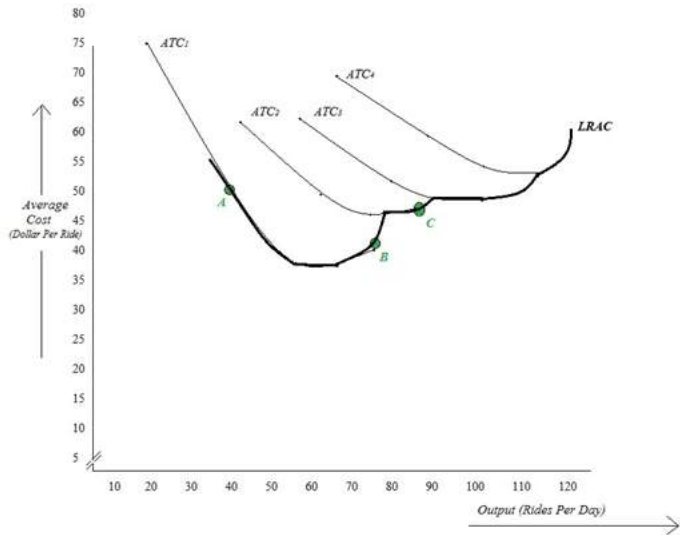


Figure-1

The points in green colors are the average cost. Points A, B and C, are showing the data for average cost against the level of output.

Therefore from the above figure, point A represents the average cost of production for 40 rides per day and is \$50, point B represents the average cost of production for 75 rides per day and is \$40 and point C represents the average cost of production for 85 rides per day and is \$47.06.

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Step 5 of 6

b.

"Minimum efficient scale can be termed as the lowest level of average total cost on which maximum output is obtained."

Observing all the tables mentioned above, the conclusion comes in the table so formed for Plant 1 as follows:

For Plant 1:

Labor (workers per day)	Output (Rides Per Day For Plant 1) Q	Total Fixed Cost (TFC) In \$	AFC (=TFC/Q) In \$	Total Variable Cost (TVC) In \$	AVC (=TVC/Q) In \$	ATC ₁ (ATC= AFC+AVC) In \$
10	20	1000	50	500	25	75
20	40	1000	25	1000	25	50
30	65	1000	15	1500	23.08	38.46

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Minimum efficient scale is at 65 rides of output with average cost is \$40.75.

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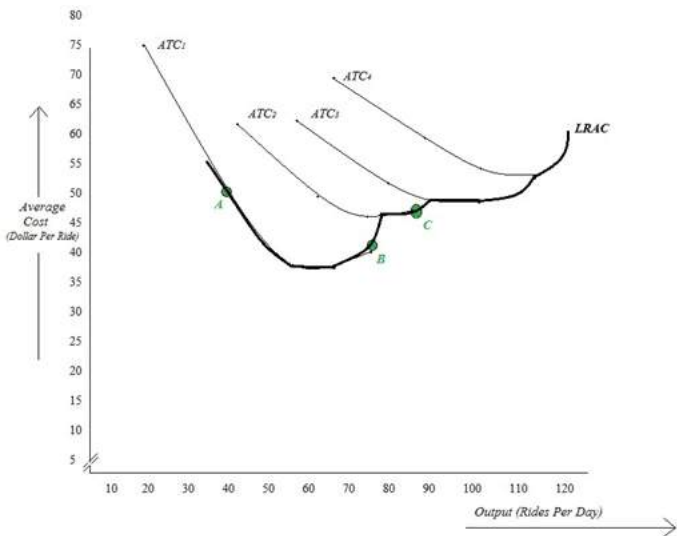
Step 6 of 6

c.

Economies of scale can be achieved by increasing the level of output.

“As the level of output increases, per unit cost of production decreases and the phenomena so called are economies of scale.”

Economies of scale can be understood with the help of the graph as follows:



It can be understood from the above graph that till the output of 65 rides per day, the average total cost is decreasing so the curve is downward sloping. Moving the output of 65 rides per day onwards, the average total cost is increasing and the graph is upwards sloping.

Therefore till 65 rides per day, production function achieves economies of scale and this point onwards it experiences diseconomies of scale.

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Following Table shows the Total Cost, Total Variable Cost, and Total Fixed Cost of Sue Surfboards at each level of output –

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- Chapter 11, Solution 12SPA

The following table shows the production function of Jackie's Canoe Rides –

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Step-by-step solution

Step 1 of 4

The short – run refers to the time period in which one factor is taken as variable that is it can be changed in order to increase or decrease the output while other factors of production are taken as fixed that is they cannot be changed in order to increase or decrease the output. The decisions in which firm or producer decides to change only one of their factors of production while keep other factors of production constant are related to the short – run and are considered as short – run decisions.

The long – run refers to the time period in which all factors of production are considered as variable and no factor is taken as fixed. This implies that in long – run in order to increase or decrease the output, all factors of production could be changed. Those decisions in which firm or producer decides to change all its factors of production are related to the long – run and are considered as long – run decisions.

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Step 2 of 4

(a) In the given news clip it has been stated that Avalon is planning to hire 15 more workers. This indicates that Avalon is increasing only one factor of production that is labor while other factors of production remain fixed. So, this decision of Avalon is a short – term decision.

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Step 3 of 4

In the given news clip it has been stated that Avalon is planning to move to a larger space. There would be increase in all factors of production such as labor, land, and capital due to this decision of the Avalon. As all factors of production are increasing, this decision of Avalon is a long run decision.

[Comment](#)

Step 4 of 4

(b) Avalon's long – run decision is riskier than the short – run decision because long – run decisions are very difficult to reverse but short run decisions could be easily reversed.

Avalon can easily reverse its short run decision by firing the additional employees it hires but reversal of long – run decision would not be easy because by moving out to a larger space Avalon will incur substantial expenditure and as cost involved is high reversing this long run decision would be very difficult and would take much longer to do so. Thus, Avalon will have to live with it for sometime even though the decision may not turn out to be right.

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Chapter 11, Solution 8SPA

Following Table shows the Total Cost, Total Variable Cost, and Total Fixed Cost of Sue Surfboards at each level of output –

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Chapter 11, Solution 12SPA

The following table shows the production function of Jackie's Canoe Rides –

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Step-by-step solution

Step 1 of 2

(a) Sunk costs are those expenditures incurred by a firm that cannot be recovered. In the same sense, the expenditure on the tickets is also a sunk cost because at the time of game expenditure incurred on the tickets could not be recovered.

I have only two options either to watch the game or not but in no circumstances I would be able to get my money back. In other words, the current opportunity cost of these tickets is zero because these tickets do not have any alternative use and any expenditure that is incurred on any good or service that does not have any alternative use is regarded as sunk cost. So, in terms of opportunity cost also expenditure on tickets, at the time of game, is sunk cost.

[Comment](#)

Step 2 of 2

(b) Cost of ticket has already been incurred and could not be recovered at all that is it has become sunk cost. This cost has to be borne by me irrespective of whether I stay at home or go to the game. Thus, for me the cost of ticket remains same regardless of whether I attend the game or not.

So, the cost of ticket would be irrelevant to my current decision. My current decision would be more influenced by the weather conditions rather than the cost of the tickets.

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Step-by-step solution

Step 1 of 8

The marginal product (MP) and average product (AP) schedule, is constructed with the help of the TP schedule.

It is given that with 1 worker, 1,000 roses a week were produced. Hiring 2nd worker doubles the total product, so with 2 workers 2,000 roses per week were produced. Hiring a 3rd worker again doubles the total product, so with 3 workers 4,000 roses per week were produced. Hiring 4th worker increases the total product by only 1,000 roses. So, with 4 workers 5,000 roses a week were produced.

[Comment](#)

Step 2 of 8

Following table shows the TP schedule based on above explanation –

[Comment](#)

Step 3 of 8

Labor	Total product (Output)
(Number of workers)	(roses per week)
1	1,000
2	2,000
3	4,000
4	5,000

Average product of labor is equal to total product divide by the quantity of labor employed.

$$\text{Average product} = \frac{\text{Total product}}{\text{Number of workers}}$$

When farm employs 1 worker, total product is 1,000 roses per week. So, average product would be:

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When farm employs 2 workers, total product is 2,000 roses per week. So, average product would be

$$\begin{aligned} \text{Average product} &= \frac{\text{Total product}}{\text{Number of workers}} \\ \text{Average product} &= \frac{2,000}{2} \\ &= 1,000 \text{ roses per week} \end{aligned}$$

[Comment](#)

Step 4 of 8

When farm employs 3 workers, total product is 4,000 roses per week. So, average product would be:

$$\begin{aligned} \text{Average product} &= \frac{\text{Total product}}{\text{Number of workers}} \\ \text{Average product} &= \frac{4,000}{3} \\ &= 1,333.33 \text{ roses per week} \end{aligned}$$

[Comment](#)

Step 5 of 8

When farm employs 4 workers, total product is 5,000 roses per week. So, average product would be:

$$\begin{aligned} \text{Average product} &= \frac{\text{Total product}}{\text{Number of workers}} \\ \text{Average product} &= \frac{5,000}{4} \\ &= 1,250 \text{ roses per week} \end{aligned}$$

Marginal product of labor is the increase in total product when quantity of labor employed is increased by one unit while other factors of production remain fixed.

For example, when rose farm increase the number of workers from 1 to 2, total product increases from 1,000 roses per week to 2,000 roses per week. So, marginal product of 2nd worker is 1,000 roses per week.

When rose farm increase the number of workers from 2 to 3, total product increases from 2,000 roses per week to 4,000 roses per week. So, marginal product of 3rd worker is 2,000 roses per week.

When rose farm increase the number of workers from 3 to 4, total product increases from 4,000 roses per week to 5,000 roses per week. So, marginal product of 4th worker is 1,000 roses per week.

[Comment](#)

Step 6 of 8

Average product and Marginal product schedule of the rose farm is as follows –

[Comment](#)

Chapter 11, Problem 17APA		Bookmark		Show all steps: ON	
Number of workers	(roses per week)	(roses per week)	(roses per week)		
1	1,000	1,000			
			1,000		
2	2,000	1,000			
			2,000		
3	4,000	1,333			
			1,000		
4	5,000	1,250			

[Comment](#)

Step 8 of 8

Increasing marginal returns occur when marginal product of an additional worker exceeds the marginal product of the previous worker.

As above table shows that up to the employment of 3 workers marginal product is increasing at each successive addition of worker to the labor force but on the employment of 4th worker marginal product is decreasing.

So, up to the employment of 3 workers, increasing marginal returns are occurring to the rose Farm run by Terri.

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Following Table shows the Total Cost, Total Variable Cost, and Total Fixed Cost of Sue Surfboards at each level of output –

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Chapter 11, Solution 12SPA

The following table shows the production function of Jackie's Canoe Rides –

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Step-by-step solution

Step 1 of 9

Following figure shows the cost data of Bill's bakery –

Total Product	AFC	AVC	ATC	MC
	(dollars per unit)	(dollars per unit)	(dollars per unit)	(dollars per unit)
10	120	100	220	
				80
20	A	B	150	
				90
30	40	90	130	
				130
40	30	C	D	
				E
50	24	108	132	

[Comment](#)

Step 2 of 9

When total product or output is 10 units, average fixed cost (AFC) is \$120 per unit.

Average fixed cost (AFC) is total fixed cost per unit of output.

$$AFC = \frac{TFC}{Q}$$

Rearranging the above equation,

$$TFC = AFC \times Q$$

$$TFC = \$120 \times 10$$

$$TFC = \$1,200$$

A is the average fixed cost when total product or output is 20 units.

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The Average Fixed Cost identified by space A is \$60 per unit.

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Step 3 of 9

B is the average variable cost when total product or output is 20 units.

AFC, when total product is 20 units, is \$60 per unit.

ATC, when total product is 20 units, is \$150 per unit.

$$ATC = AFC + AVC$$

Rearranging the above equation,

$$AVC = ATC - AFC$$

$$AVC = \$150 \text{ per unit} - \$60 \text{ per unit}$$

$$AVC = \$90 \text{ per unit.}$$

The Average Variable Cost identified by space B is \$90 per unit.

[Comment](#)

Step 4 of 9

When 30 units are produced, average total cost (ATC) is \$130 per unit.

Average total cost (ATC) is total cost per unit of output.

$$ATC = \frac{TC}{Q}$$

Rearranging above equation,

$$TC = ATC \times Q$$

$$TC = \$130 \text{ per unit} \times 30$$

$$TC = \$3,900$$

Total cost when 30 units are produced is \$3,900

Marginal cost is the change in total cost divided by change in total output.

$$MC = \frac{\Delta TC}{\Delta Q}$$

Rearranging above equation,

$$?TC = MC \times ?Q$$

When output increases from 30 units to 40 units, marginal cost (MC) is \$130 per unit

$$?Q = 10 \text{ units}$$

$$MC = \$130 \text{ per unit}$$

$$?TC = MC \times ?Q$$

[Comment](#)

Step 5 of 9

$$= \$130 \times 10$$

$$= \$1,300$$

So, increase in total cost when output increases from 30 units to 40 units = \$1300

Chapter 11, Problem 19APA

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D is the average total cost when 40 units are produced.

Total cost (TC) when 40 units are produced = \$5,200

$$ATC = \frac{TC}{Q}$$

$$ATC = \frac{\$5,200}{40}$$

$$= \$130$$

The Average Total Cost identified by space D is \$130 per unit.

[Comment](#)

Step 6 of 9

C is the average variable cost when total product or output is 40 units.

AFC, when total product is 40 units, is \$30 per unit.

ATC, when total product is 40 units, is \$130 per unit.

$$ATC = AFC + AVC$$

Rearranging the above equation,

$$AVC = ATC - AFC$$

$$AVC = \$130 \text{ per unit} - \$30 \text{ per unit}$$

$$AVC = \$100 \text{ per unit.}$$

The Average Variable Cost identified by space C is \$100 per unit.

[Comment](#)

Step 7 of 9

E is the marginal cost when output increases from 40 units to 50 units.

Marginal cost (MC) is equal to the change in total cost divided by change in total output.

$$MC = \frac{\Delta TC}{\Delta Q}$$

For calculating the marginal cost, we need to have total cost when output is 40 units and 50 units respectively.

When output is 40 units, ATC is \$130 per unit

$$ATC = \frac{TC}{Q}$$

Rearranging above equation,

$$TC = ATC \times Q$$

$$TC = \$130 \text{ per unit} \times 40 \text{ units}$$

$$TC = \$5,200$$

Total cost when 40 units are produced is \$5,200

[Comment](#)

Step 8 of 9

When output is 50 units, ATC is \$132 per unit

Chapter 11, Problem 19APA

4 Bookmarks

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TC = \$132 per unit × 50 units

TC = \$6,600

Total cost when 50 units are produced is \$6,600.

[Comment](#)

Step 9 of 9

Change in Total cost when output increases from 40 to 50 units = Total cost when 50 units are produced – Total cost when 40 units are produced.

Change in Total cost when output increases from 40 to 50 units = \$6,600 - \$5,200

= \$1,400

$$MC = \frac{\Delta TC}{\Delta Q}$$

$$= \frac{\$1,400}{10}$$

= \$140 per unit

The Marginal Cost identified by space E is \$140 per unit

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Chapter 11, Solution 8SPA

Following Table shows the Total Cost, Total Variable Cost, and Total Fixed Cost of Sue Surfboards at each level of output –

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Chapter 11, Solution 12SPA

The following table shows the production function of Jackie's Canoe Rides –

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Step-by-step solution

Step 1 of 16

ProPainters hires students at \$250 a week to paint houses. It leases equipment at \$500 a week.

[Comment](#)

Step 2 of 16

Following table shows the Total Product schedule of the ProPainters –

[Comment](#)

Step 3 of 16

Labor	Output
(students)	(houses painted per week)
1	2
2	5
3	9
4	12
5	14
6	15

[Comment](#)

Step 4 of 16

ProPainters leases equipment at \$500 a week. Leasing of equipment is a fixed cost for the ProPainters.

So, its total fixed cost is \$500 per week.

ProPainters hire students at \$250 a week to paint houses. Hiring of students is a variable cost of the ProPainters. Its variable cost depends on the number of students it hires.

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= \$1,000 per week

TC = TFC + TVC

T_L = \$500 per week + \$1,000 per week

TC = \$1,500 per week

The Total Cost of ProPainters when they paint 12 houses is \$1,500 per week.

[Comment](#)

Step 5 of 16

Average total cost is total cost per unit of output.

When ProPainters paint 12 houses a week, its Total Cost is \$1,500 per week.

$$\begin{aligned}
 ATC &= \frac{TC}{Q} \\
 &= \frac{\$1,500}{12} \\
 &= \$125 \text{ per house}
 \end{aligned}$$

The Average Total Cost of ProPainters when they paint 12 houses is \$125 per house.

[Comment](#)

Step 6 of 16

Marginal cost is the increase in total cost divided by the increase in output.

$$MC = \frac{\Delta TC}{\Delta Q}$$

In order to calculate marginal cost of ProPainters when they paint 12 houses, we have to calculate the total cost when they paint 9 houses and 12 houses a week respectively.

When ProPainters paint 9 houses a week, they hire 3 workers

So,

$$\begin{aligned}
 TVC &= 3 \times \$250 \\
 &= \$750 \text{ per week} \\
 TFC &= \$500 \text{ per week} \\
 TC &= TFC + TVC \\
 TC &= \$500 + \$750 \\
 &= \$1,250 \text{ per week}
 \end{aligned}$$

[Comment](#)

Step 7 of 16

When ProPainters paint 12 houses, their total cost is \$1,500 per week (calculated above).

Change in total cost when number of houses painted a week increase from 9 to 12 = Total cost of painting 12 houses a week – Total cost of painting 9 houses a week

Change in total cost when number of houses painted a week increase from 9 to 12 = \$1,500 - \$1,250

Change in total cost when number of houses painted a week increase from 9 to 12 = \$250

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$$MC = \$83.33 \text{ per house}$$

T : Marginal Cost of ProPainters when they paint 12 houses is \$83.33 per house.

To ascertain at what output, ATC is minimum we have to calculate the ATC at all output.

For example, when ProPainters paints 2 houses a week, they hire only 1 student. So,

$$TVC = \$250 \text{ per week}$$

$$TFC = \$500 \text{ per week}$$

$$TC = TFC + TVC$$

$$TC = \$500 + \$250$$

$$= \$750$$

$$ATC = \frac{TC}{Q}$$

$$ATC = \frac{\$750}{2}$$

$$ATC = \$375 \text{ per house}$$

[Comment](#)

Step 8 of 16

When ProPainters paints 5 houses a week, they hire only 2 students. So,

$$TVC = \$250 \times 2$$

$$= \$500 \text{ per week}$$

$$TFC = \$500 \text{ per week}$$

$$TC = TFC + TVC$$

$$TC = \$500 + \$500$$

$$= \$1,000$$

$$ATC = \frac{TC}{Q}$$

$$ATC = \frac{\$1,000}{5}$$

$$ATC = \$200 \text{ per house}$$

[Comment](#)

Step 9 of 16

When ProPainters paints 9 houses a week, they hire only 3 students. So,

$$TVC = \$250 \times 3$$

$$= \$750 \text{ per week}$$

$$TFC = \$500 \text{ per week}$$

$$TC = TFC + TVC$$

$$TC = \$500 + \$750$$

$$= \$1,250$$

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Step 10 of 16

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$$ATC = \$138.88 \text{ per house}$$

[Comment](#)

Step 11 of 16

When ProPainters paints 12 houses a week, they hire only 4 students. So,

$$\begin{aligned} TVC &= \$250 \times 4 \\ &= \$1,000 \text{ per week} \\ TFC &= \$500 \text{ per week} \\ TC &= TFC + TVC \\ TC &= \$500 + \$1,000 \\ &= \$1,500 \end{aligned}$$

[Comment](#)

Step 12 of 16

$$\begin{aligned} ATC &= \frac{TC}{Q} \\ ATC &= \frac{\$1,500}{12} \\ ATC &= \$125 \text{ per house} \end{aligned}$$

[Comment](#)

Step 13 of 16

When ProPainters paints 14 houses a week, they hire only 5 students. So,

$$\begin{aligned} TVC &= \$250 \times 5 \\ &= \$1,250 \text{ per week} \\ TFC &= \$500 \text{ per week} \\ TC &= TFC + TVC \\ TC &= \$500 + \$1,250 \\ &= \$1,750 \end{aligned}$$

[Comment](#)

Step 14 of 16

$$\begin{aligned} ATC &= \frac{TC}{Q} \\ ATC &= \frac{\$1,750}{14} \\ ATC &= \$125 \text{ per house} \end{aligned}$$

When ProPainters paints 15 houses a week, they hire only 6 students. So,

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= \$2,000

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Step 15 of 16

$$ATC = \frac{TC}{Q}$$

$$ATC = \frac{\$2,000}{15}$$

$$ATC = \$133.33 \text{ per house}$$

Tabulating the above calculated values of Average total cost of ProPainters –

Output	Average Total Cost (ATC)
(houses painted per week)	(dollars per house)
2	375
5	200
9	138.88
12	125
14	125
15	133.33

[Comment](#)

Step 16 of 16

As above table shows that ProPainters will have the minimum average total cost when it paint between 12 to 14 houses per week that is when it paints 13 houses per week.

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Chapter 11, Problem 25APA

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Step-by-step solution

Step 1 of 9

The following table shows the production function of Bonnie's Balloon Rides –

Labor (workers per day)	Output (rides per day)	
	Plant 1	Plant 2
10	6	10
20	10	15
30	13	18
40	15	20
50	16	21
Balloons (numbers)	1	2

Comment

Step 2 of 9

Rent of balloon is \$500 a day. This is the fixed cost for Bonnie's Balloon Rides.

Each balloon operator is paid \$25 a day. This is the variable cost for Bonnie's Balloon Rides.

For example, if Bonnie's operates plant 1 with 1 balloon and hires 10 workers, it undertakes 6 rides. Its total cost is as follows –

Total fixed cost = 1 × \$500

= \$500

Total variable cost = \$25 × 10

= \$250

Comment

Step 3 of 9

Total cost = Total variable cost + Total fixed cost

= \$250 + \$500

= \$750

Comment

Step 4 of 9

Average total cost is the total cost divided by quantity produced. As above calculation shows that for undertaking 6 rides, Bonnie's is incurring total cost of \$750. So,

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$$ATC = \frac{1250}{10}$$
$$= \$125 \text{ per ride}$$

Similarly, ATC for other level of output (number of rides undertaken per day) could be calculated.

Comment

Step 5 of 9

Now, if Bonnie's operates plant 2 with 2 balloons and hires 10 workers, it undertakes 10 rides. Its total cost is as follows –

Total fixed cost = 2 × \$500

= \$1,000

Total variable cost = \$25 × 10

= \$250

Total cost = Total variable cost + Total fixed cost

= \$250 + \$1,000

= \$1,250

Average total cost is the total cost divided by quantity produced. As above calculation shows that for undertaking 10 rides, Bonnie's is incurring total cost of \$1,250. So,

$$ATC = \frac{TC}{Q}$$
$$ATC = \frac{\$1,250}{10}$$
$$= \$125 \text{ per ride}$$

Similarly, ATC for other level of output (number of rides undertaken per day) could be calculated.

Comment

Step 6 of 9

Following table shows the ATC when Bonnie's operates Plant 1 –

Output (rides per day)	Average Total Cost (ATC) (dollars per ride)
6	125
10	100
13	96.15
15	100
16	109.38

Comment

Step 7 of 9

Following table shows the ATC when Bonnie's operates Plant 2 –

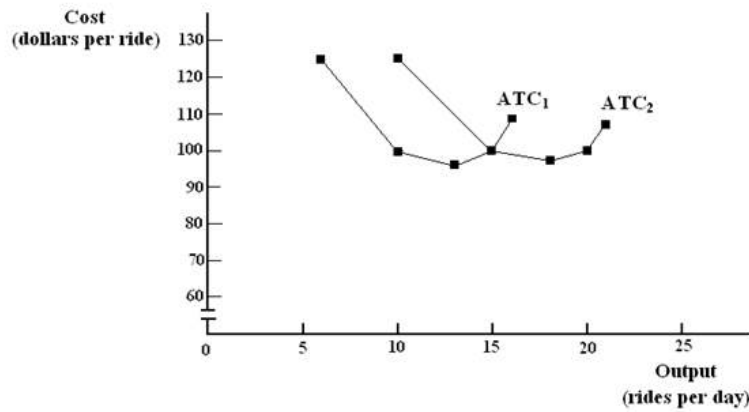
Output (rides per day)	Average Total Cost (ATC) (dollars per ride)
10	125
15	100
18	97.22
20	100
21	107.14

Following figure shows the average total cost curve of Bonnie's Balloon Rides when it operate Plant 1 and Plant 2 –

Comment



Step 8 of 9

[Comment](#)

Step 9 of 9

In the above figure output is taken on the X-axis and average total cost (ATC) is taken on Y – axis. ATC_1 represents average total cost when Bonnie's operate Plant 1 and ATC_2 represents average total cost when Bonnie's operate Plant 2.

These ATC curves differ because they both pertain to different plants. ATC_1 represents Plant 1 with 1 balloon while ATC_2 represents Plant 2 with 2 balloons.

Secondly, for each short – run ATC curve, the larger the plant, the greater is the output at which average total cost is at minimum. As above figure shows that for smaller Plant 1, ATC is minimum when 13 rides per day is conducted while for larger plant 2, ATC is minimum when 18 rides per day is conducted.

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Chapter 11, Solution 8SPA

Following Table shows the Total Cost, Total Variable Cost, and Total Fixed Cost of Sue Surfboards at each level of output –

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The following table shows the production function of Jackie's Canoe Rides –

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Step-by-step solution

Step 1 of 10

The following table shows the production function of Bonnie's Balloon Rides –

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Step 2 of 10

Labor (workers per day)	Output (rides per day)	
	Plant 1	Plant 2
10	13	15
20	18	20
30	22	24
40	24	26
50	25	27
Balloons (numbers)	3	4

[Comment](#)

Step 3 of 10

Rent of balloon is \$500 a day. This is the fixed cost for Bonnie's Balloon Rides.

Each balloon operator is paid \$25 a day. This is the variable cost for Bonnie's Balloon Rides.

For example, if Bonnie's operates Plant 3 with 3 balloons and hires 10 workers, it undertakes 13 rides. Its total cost is as follows –

Total fixed cost = 3 × \$500

= \$1,500

Total variable cost = \$25 × 10

= \$250

Total cost = Total variable cost + Total fixed cost

= \$250 + \$1,500

= \$1,750

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Step 4 of 10

Average total cost is the total cost divided by quantity produced. As above calculation shows that for undertaking 13 rides, Bonnie's is incurring total cost of \$1,750. So,

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Similarly, ATC for other level of output (number of rides undertaken per day) could be calculated.

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Step 5 of 10

Now, if Bonnie's operates Plant 4 with 4 balloons and hires 10 workers, it undertakes 15 rides. Its total cost is as follows –

Total fixed cost = 4 × \$500

= \$2,000

Total variable cost = \$25 × 10

= \$250

Total cost = Total variable cost + Total fixed cost

= \$250 + \$2,000

= \$2,250

Comment

Step 6 of 10

Average total cost is the total cost divided by quantity produced. As above calculation shows that for undertaking 15 rides, Bonnie's is incurring total cost of \$2,250. So,

ATC = TC / Q

ATC = \$2,250 / 15

= \$150 per ride

Similarly, ATC for other level of output (number of rides undertaken per day) could be calculated.

Comment

Step 7 of 10

Following table shows the ATC when Bonnie's operates Plant 3 –

Output (rides per day)	Average Total Cost (ATC) (dollars per ride)
13	134.61
18	111.11
22	102.27
24	104.17
25	110

Comment

Step 8 of 10

Following table shows the ATC when Bonnie's operates Plant 4 –

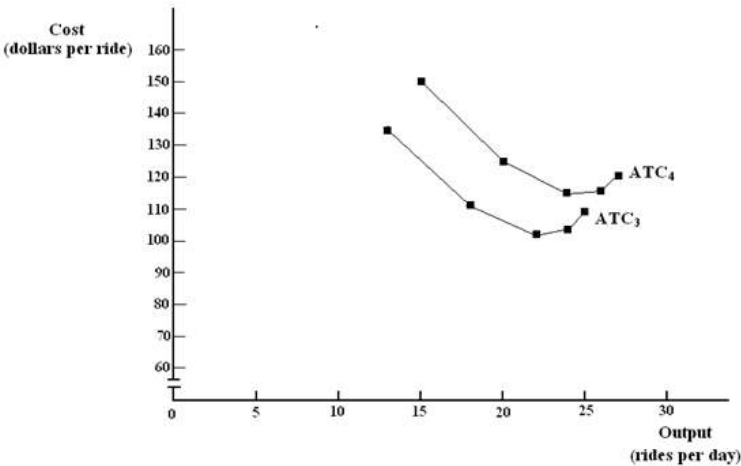
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26	115.38
257	120.37

Following figure shows the average total cost curve of Bonnie's Balloon Rides when it operate Plant 3 and Plant 4 –

Comment

Step 9 of 10



Comment

Step 10 of 10

In the above figure output is taken on the X-axis and average total cost (ATC) is taken on Y – axis. ATC₃ represents average total cost when Bonnie's operate Plant 3 and ATC₄ represents average total cost when Bonnie's operate Plant 4.

These ATC curves differ because they both pertain to different plants. ATC₃ represents Plant 3 with 3 balloons while ATC₄ represents Plant 4 with 4 balloons.

Secondly, for each short – run ATC curve, the larger the plant, the greater is the output at which average total cost is at minimum. As above figure shows that for smaller Plant 3, ATC is minimum when 22 rides per day is conducted while for larger plant 4, ATC is minimum when 24 rides per day is conducted.

Comment

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1

Recommended solutions for you in Chapter 11

Chapter 11, Solution 8SPA

Following Table shows the Total Cost, Total Variable Cost, and Total Fixed Cost of Sue Surfboards at each level of output –

Chapter 11, Solution 12SPA

The following table shows the production function of Jackie's Canoe Rides –

[See solution](#)

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Step-by-step solution

Step 1 of 5

“LRAC curve tells the minimum efficient scale of point on which firm should operate.”

In order to draw the LRAC curve, ATC curves for different output has to be drawn first. Therefore calculation for ATC curve is as follows:

For Plant 1:

Labor (workers per day)	Output (Rides Per Day For Plant 1) Q	Total Fixed Cost (TFC) In \$	AFC (=TFC/Q) In \$	Total Variable Cost (TVC) In \$	AVC (=TVC/Q) In \$	ATC ₁ (ATC= AFC+AVC) In \$
10	6	500	83.33	250	41.67	125
20	10	500	50	500	50	100
30	13	500	38.46	750	57.69	96.15
40	15	500	33.33	1000	66.67	100
50	16	500	31.25	1250	78.13	109.38

Comment

Step 2 of 5

Cost of balloon ride is \$500 per day and the cost of labor is \$25 per day per labor. Since for first plant number total 1 balloon is used and the cost of each of the Canoe is \$100 so the fixed cost will be as follows:

Total Fixed Cost = Cost Per Balloon × Total Number of Balloon
Total Fixed Cost = 500 × (1)
Total Fixed Cost = \$500

Similarly the Total Variable cost is as follows:

Total Variable Cost = Cost Per Lobor × Total number of Labors
Total Variable Cost = 25 × 10
Total Variable Cost = \$250

When there are 10 labors deployed in the operation, the total variable cost is coming out to be \$250 the same calculation is done for different number of labors deployed in the operation.

The calculation for AVC, AFC and ATC₁ is itself explained in the table. As a result of all the calculation data of ATC₁ at different level of output is found.

Similarly calculation for **Plant 2**, **Plant 3** and **Plant 4** is done and the data so obtained is found in the following three tables as follows:

Comment

Step 3 of 5

For Plant 2:

Labor (workers per day)	Output (Rides Per Day For Plant 2) Q	Total Fixed Cost (TFC) In \$	AFC (=TFC/Q) In \$	Total Variable Cost (TVC) In \$	AVC (=TVC/Q) In \$	ATC ₂ (ATC= AFC+AVC) In \$

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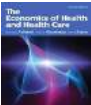
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20	15	1000	66.67	500	33.33	100
30	18	1000	55.56	750	41.67	97.22
40	20	1000	50	1000	50	100
50	21	1000	47.62	1250	59.52	107.14

For Plant 3:

Labor (workers per day)	Output (Rides Per Day For Plant 3) Q	Total Fixed Cost (TFC) In \$	AFC (=TFC/Q) In \$	Total Variable Cost (TVC) In \$	AVC (=TVC/Q) In \$	ATC ₂ (ATC= AFC+AVC) In \$
10	13	1500	115.38	250	19.23	134.62
20	18	1500	83.33	500	27.78	111.11
30	22	1500	68.18	750	34.09	102.27
40	24	1500	62.50	1000	41.67	104.17
50	25	1500	60	1250	50	110

For Plant 4:

Labor (workers per day)	Output (Rides Per Day For Plant 4) Q	Total Fixed Cost (TFC) In \$	AFC (=TFC/Q) In \$	Total Variable Cost (TVC) In \$	AVC (=TVC/Q) In \$	ATC ₄ (ATC= AFC+AVC) In \$
10	15	2000	133.33	250	16.67	150.00
20	20	2000	100	500	25	125.00
30	24	2000	83.33	750	31.25	114.58
40	26	2000	76.92	1000	38.46	115.38
50	27	2000	74.07	1250	46.29	120.36

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Step 4 of 5

(i)

On the basis of above data, ATC curves for all the plants are drawn and LRAC curve is going through the minimum point of all the ATC curves. Figure 1 is representing the graphical form of data and is shown as follows:

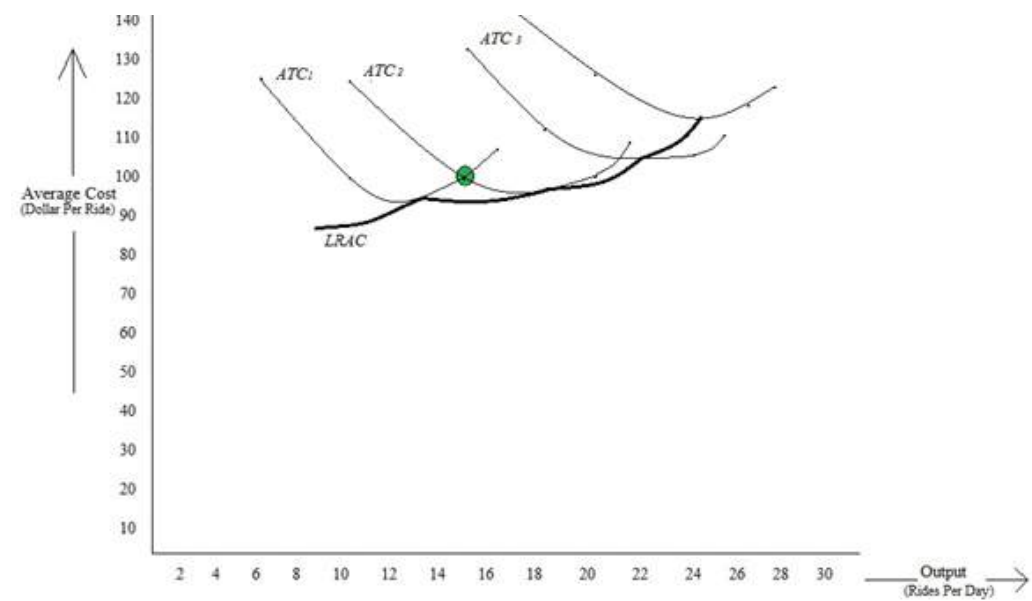


Figure-1

The green point is showing the data for average cost against the level of output.

Therefore from the above figure, the point represents the average cost of production for 15 rides per day and the cost is \$100.

Therefore average cost of production for 15 rides per day is \$100.

[Comment](#)

Step 5 of 5

(ii)

On the basis of above data, ATC curves for all the plants are drawn and LRAC curve is going through the minimum point of all the ATC curves. Figure 2 is representing the graphical form of data and is shown as follows:

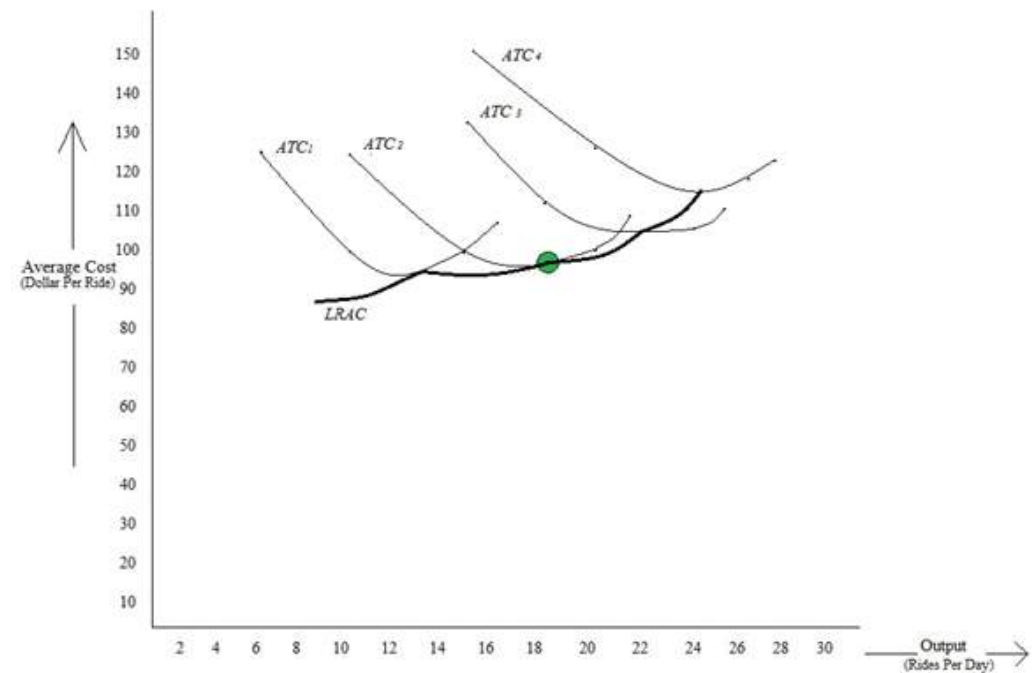


Figure-2

The green point is showing the data for average cost against the level of output.

Therefore from the above figure, the point represents the average cost of production for 18 rides per day and the cost is \$97.22.

Therefore average cost of production for 15 rides per day is \$97.22.

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Recommended solutions for you in Chapter 11

Chapter 11, Solution 8SPA

Following Table shows the Total Cost, Total Variable Cost, and Total Fixed Cost of Sue Surfboards at each level of output –

[See solution](#)

Chapter 11, Solution 12SPA

The following table shows the production function of Jackie's Canoe Rides –

[See solution](#)

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Step-by-step solution

Step 1 of 4

a.

Cost of kiosk is \$40,000. The cost itself is the fixed cost for the machine that is being used. Coinstar is the machine used for getting cash by depositing the coins into the machine. The variable cost is the cost of bean that is being used, the cost of the outlet premise etc. Machine that is being used need the maintenance if used frequently. Maintenance of kiosk is also the variable cost.

Comment

Step 2 of 4

b.

As the cost of barista operated machine costs less, the fixed cost of barista operated machine decreases but what had been the variable cost for the barista operated machine tends to increase. On the other hand the kiosk machine will be at \$40,000 which will be greater than that of barista operated machine. Therefore the fixed cost of the coinstar will increase. The total cost of coinstar will now be greater in case of operation at lower levels. Total cost will be lower if self serviced kiosk is operated at higher levels.

Comment

Step 3 of 4

c.

Table representing the data for Barista operated machines when fixed cost is less than \$40,000. The cost is now \$30,000 (on assumption) and the table is as follows:

	Labor	Output	Fixed Cost	Variable Cost	Total Cost	Marginal Cost	ATC
A	0	0	30000	-	30000	-	-
B	10	400	30000	10000	40000	25	100
C	20	1000	30000	18000	48000	13.33	48.00
D	30	1300	30000	26000	56000	26.67	43.08
E	40	1500	30000	34000	64000	40	42.67
F	50	1600	30000	42000	72000	80	45

The formula for finding the marginal cost is as follows:

Marginal Cost = $\frac{\Delta TC}{\Delta \text{Output}}$

Marginal Cost = $\frac{40000 - 30000}{400 - 0}$

Marginal Cost = $\frac{10000}{400} = 25$

Above marginal cost is found at the output level of 400 units. Similarly for the consecutive output levels, marginal cost is found.

Average Total cost is found by division of Total cost by output.

Table representing the data for Coinstar machine having the cost of \$40,000 is as follows:

	Machines	Output	Fixed Cost	Variable Cost	Total Cost	Marginal Cost	ATC

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C	20	1000	40000	22000	62000	11.67	62.00
	30	1300	40000	25000	65000	10.00	50.00
E	40	1500	40000	32000	72000	35	48.00
F	50	1600	40000	39000	79000	70	49.375

The formula for finding the marginal cost is as follows:

Marginal Cost = $\frac{\Delta TC}{\Delta \text{Output}}$

Marginal Cost = $\frac{55000 - 40000}{400 - 0}$

Marginal Cost = $\frac{15,000}{400} = 37.5$

Above marginal cost is found at the output level of 400 units. Similarly for the consecutive output levels, marginal cost is found.

Average Total cost is found by division of Total cost by output.

Figure-3 representing the conditions as explained in the above tables is as follows:

[Comment](#)

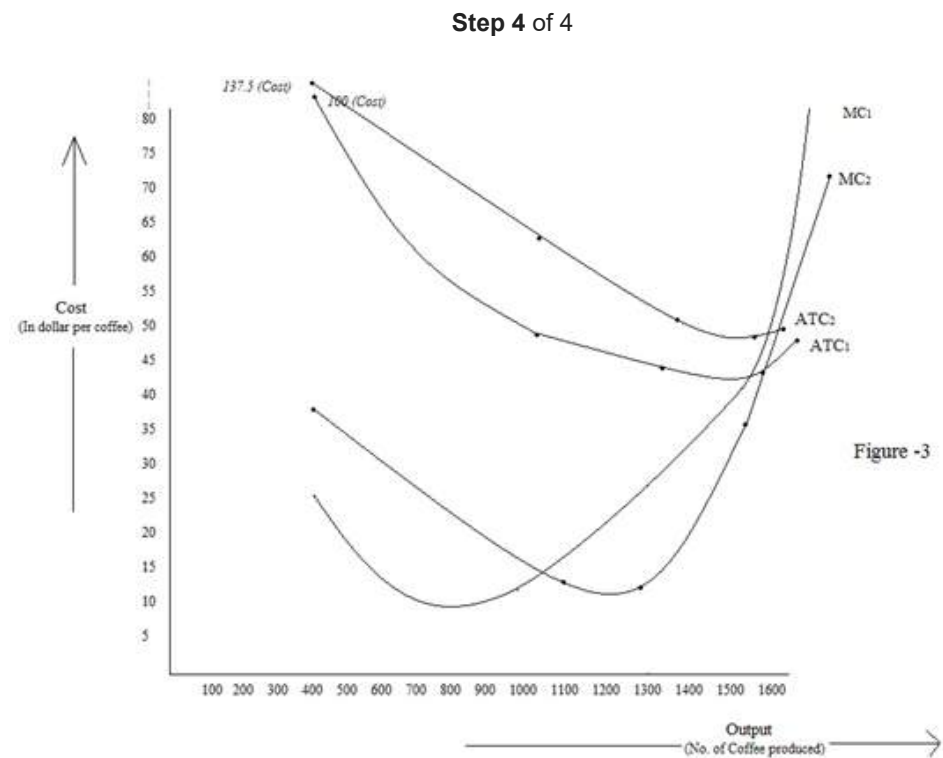


Figure-3 shows the marginal cost curve and total average cost curve for barista operated machines and for coinstar machines. The costs with the barista-operated machine are labeled “1” and the costs with the Coinstar are labeled “2”.

The average total cost of the Coinstar machine is higher than that of the barista-operated machine at low levels of output and is lower than the average total cost of the barista-operated machine at high levels of output. The marginal cost of the Coinstar machine is lower than the marginal cost of the barista-operated machine.

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Recommended solutions for you in Chapter 11

Chapter 11, Solution 8SPA

Following Table shows the Total Cost, Total Variable Cost, and Total Fixed Cost of Sue Surfboards at each level of output –

[See solution](#)

Chapter 11, Solution 12SPA

The following table shows the production function of Jackie’s Canoe Rides –

[See solution](#)

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